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Daniel Egbo

Curriculum Vitae

Education

April 2021 - July 2026 **PhD, Astronomy (Thesis Submitted)**, *University of Cape Town*, Cape Town,
Thesis: Multiwavelength Study of the SARAO MeerKAT Galactic Plane Point Sources
Advisors: Prof. David Buckley and Prof. Paul Groot

Dec. 2019 **MSc. in Physics (Astrophysics optional)**, *University of Nigeria*, Nsukka, *Grade*
Thesis: Photometric Study of White Dwarf Pulsar AR Scorpii
Advisors: Dr. F. I. Odoh

Aug. 2014 **BSc. (Hons) Physics & Astronomy**, *University of Nigeria*, Nsukka,

Research Interests

- Variable stars: chromospherically active stars, radio stars, flare stars, X-ray stars, young stellar objects, transients, spectroscopy and photometry in observational astrophysics.
- Stellar Planet Interactions: brown dwarfs, exoplanets
- Statistical/Machine Learning: multi-wavelength cross-correlation, regression, classification, clustering, anomaly detection, deep learning, Generative AI research

Conferences & Seminar Talks

2025 The 5th Annual AfAS Conference, UNISA, Johannesburg, South Africa, contributed talk: "Corona and Chromospheric Active Stars from the SARAO MeerKAT Galactic Plane Survey

2024 International Astronomical Union General Assembly, Division G-3: Stars and Stellar Physics, Cape Town, South Africa, poster: "A Multi-wavelength Study of the Galactic Point Sources in the MeerKAT Galactic Plane Survey with GAIA DR3"

2024 The 4th Annual AfAS Conference, Marrakech, Morocco, contributed talk: "A Multi-wavelength Study of the Galactic Point Sources in the MeerKAT Galactic Plane Survey with GAIA DR3"

2024 MeerKAT@5 Conference, Stellenbosch, South Africa, poster: "A Multi-wavelength Study of the Galactic Point Sources in the MeerKAT Galactic Plane Survey with GAIA DR3"

2023 Surveying the Milky Way: The Universe in Our Own Backyard, Pasadena, California, US, contributed talk: "A Multi-wavelength Study of the Galactic Point Sources in the MeerKAT Galactic Plane Survey with GAIA DR3"

2023 2023 ThunderKAT Workshop, University of Oxford, UK, contributed talk: "A Multi-wavelength Study of the Galactic Point Sources in the MeerKAT Galactic Plane Survey with GAIA DR3"

2023 The Milky Way Revealed by Gaia: The Next Frontier, University of Barcelona, Spain, contributed talk: "A Multi-wavelength Study of the Galactic Point Sources in the MeerKAT Galactic Plane Survey with GAIA DR3"

Specialised Trainings/Workshops

August 2026	2026 IAIFI Summer School
June 2026	2026 Harvard AstroAI Workshop 2026
March 2026	AI in Radio Astronomy Workshop: Preparing the UK for the SKA Data Future, Edinburgh, Scotland
August 2024	LSST Workshop at the IAU GA, Cape Town, South Africa
May 2023	Oxford Machine Learning (OxML) Summer School
Feb 2023	COSPAR X-ray Vision School of the Energetic Universe, North West University, Potchefstroom, South Africa
August 2021	ZTF Summer School
June 2021	ESCAPE Summer School - Data Science for Astronomy, Astroparticle and Particle Physics
August 2020	GROWTH Summer School

Observational Experience

2023 - 2025	South African Astronomical Observatory, SA, multiple observations using SpUPNIC spectroscopic instrument mounted on the 1.9 m Telescope
2021	South African Astronomical Observatory, SA, multiple observations using SHOC photometric instrument mounted on the 1.0 m Telescope

Volunteer Opportunities

Oct 2026	Judge, 6th Big Data Africa School- a program organised by the Development of Africa through Radio Astronomy (DARA).
May 2026 - July 2026	Lead Facilitator, 2026 BRICS Astronomy & IDIA Data Analytics Program.
May 2025 - July 2025	Lead Facilitator, 2025 BRICS Astronomy & IDIA Data Analytics Program.
Sept 2024 - March 2025	Volunteer/Mentor, 5th Big Data Africa School- a program organised by the Development of Africa through Radio Astronomy (DARA).
Aug 2022 - Present	SAAO Open Night- a bimonthly talk with SAAO tour and star gazing event at the SAAO facility in Cape Town.
Aug 2024	Volunteer, International Astronomical Union General Assembly (IAU GA), Cape Town, South Africa
Nov - Dec 2023	Observational and Data Analysis Trainer, International School of Young Astronomers (ISYA), SAAO, Cape Town, South Africa

Scholarships and Grants

March 2026	Royal Society Support Grant for Visiting The Open University, UK for Machine learning in data-intensive Astronomy
2021 - 2024	SAAO PhD Prize Scholarship
2023	Breakthrough Listen Travel Grant to attend the ThunderKAT meeting at the University of Oxford.
Sept-Nov, 2018	African German Network of Excellence in Science (AGNES) Intra-Africa Mobility Grant.

Publications

2026 **Egbo, O. D.**, Buckley, D. A. H., Groot, P. J., Mogawana, O., van Dyk, A., Tampo, Y. & de Wet, S. (2026). Chromospherically Active Stars in the MeerKAT Galactic Plane Survey. *Monthly Notices of the Royal Astronomical Society*. doi:10.1093/mnras/stag1821 (published online).

2026 **Egbo, O. D.**, Groot, P. J., Buckley, D. A. H., Robrade, J., Schwope, A. D., Freund, S., Schneider, P. C. & Stelzer, B. (2026). X-ray counterparts to stellar MeerKAT Galactic-plane compact radio sources. *Astronomy & Astrophysics*, 707, A193. doi:10.1051/0004-6361/202556983.

2026 Brink, J., Groot, Buckley, D. A. H., Veresvarska, M., Schwope, A. D., Groot, P. J., Thorstensen, J. R., Cúneo, V. A., Potter, S. B., Titus, N., **Egbo, O. D.**, Lees, R., Mogawana, O., & van Dyk, A. (2026). The optical photometric and spectroscopic periodicities of the cataclysmic variable SRGt 062340.2-265751. *Astronomy & Astrophysics*, 707, A25. doi:10.1051/0004-6361/202557005.

2025 **Egbo, O. D.**, Buckley, D. A. H., Groot, P. J., Cavallaro, F., Woudt, P. A., Thompson, M. A., Mutale, M. & Bietenholz, M. (2025). The stellar population in the SARAO MeerKAT Galactic Plane Survey. *Monthly Notices of the Royal Astronomical Society*, 540(3), pp.2685-2702. doi:10.1093/mnras/staf847.

2024 Erasmus, N., Steele, I. A., Piascik, A. S., Bates, S. D., Mottram, C. J., Rosie, K. A., ... & **Egbo, O. D.**. (2024). Mookodi: multi-purpose low-resolution spectrograph and multi-filter photometric imager for rapid follow-up observations of astronomical transient events. *Journal of Astronomical Telescopes, Instruments, and Systems*, 10(2), 025005-025005. doi:10.1117/1.JATIS.10.2.025005.

2024 Potter, S. B., Buckley, D. A., Scaringi, S., Monageng, I. M., **Egbo, O. D.**, Charles, P. A., ... & Hlakola, M. (2024). Optical spectroscopic and photometric classification of the X-ray transient EP240309a (EP J115415. 8- 501810) as an intermediate polar. *Monthly Notices of the Royal Astronomical Society: Letters*, 532(1), L21-L26. doi:10.1093/mnrasl/slae038.

2024 Goedhart, S., Cotton, W. D., Camilo, F., Thompson, M. A., Umana, G., Bietenholz, M., ... & **Egbo, O. D.** ... & Williams, L. P. (2024). The SARAO MeerKAT 1.3 GHz Galactic Plane Survey. *Monthly Notices of the Royal Astronomical Society*, 531(1), 649-681. doi:10.1093/mnras/stae1166.

2023 Garnavich, P., Potter, S. B., Buckley, D. A., van Dyk, A., **Egbo, O. D.**, Littlefield, C., & Greiveldinger, A. (2023). Rapid Evolution of the White Dwarf Pulsar AR Scorpii. *The Astrophysical Journal Letters*, 958(2), L22. doi:10.3847/2041-8213/ad0be7.

2023 O'Connor, B., Kouveliotou, C., Evans, P. A., Gorgone, N., Van Kooten, A. J., Gagnon, S., ... & **Egbo, O. D.** ... & Younes, G. (2023). The swift deep galactic plane survey (DGPS) phase I catalog. *The Astrophysical Journal Supplement Series*, 269(2), 49. doi:10.3847/1538-4365/ad0228.

Computational Skills

Programming languages	Python, Bash and R
Astronomy Data Tools	Astropy, ADQL, TOPCAT, IRAF/pyRAF, CARTA, DS9

Skill Matrix

	Level	Skill	Years	Comment
Language:	■■■■■	Python	7	*Proficient*
	■■■■■	R	4	*Intermediate*
	■■■■■	LaTeX	6	*LaTeX for my academic writing*
OS:	■■■■■	Linux	6	*I use linux and bash scripting.*
Database:	■■■■■	PostgreSQL	3	*Can create database, table and perform analytics.*
	■■■■■	Google BigQuery	3	*Creation of tables and analytics.*
Methods:	■■■■■	Numerical methods	4	*Statistical Analysis, cross-correlation*
	■■■■■	Data Analytics	6	*EDA, Visualisation, Feature engineering,*
	■■■■■	Machine Learning	5	*Regression, Classification, Clustering*
	■■■■■	MLOps	3	*Concepts of feature, training and inference (FTI) pipeline.*
	■■■■■	Deep Learning	3	*Image analysis, Diffusion, transfer learning*
	■■■■■	GPU Programming	2	*Accelerated data processing, accelerated machine learning, CUDA workflow*

Trainings

2025 Data Talks Club - MLOps Zoomcamp
2024 NVIDIA - Generative AI with Diffusion Models
2024 NVIDIA - Getting Started with Deep Learning
2024 NVIDIA - Accelerating End-to-End Data Science Workflows
2023 Data Talks Club Machine Learning ZoomCamp
2021 WorldQuant Scientific Computing & Python
2021 WorldQuant Machine Learning & Statistical Analysis.